1️⃣ Dictionary in Python

Definition

A **dictionary** is a built-in data type in Python used to store **data in key–value pairs**.

Each key is unique and maps to a value.

Dictionaries are **unordered**, **mutable (changeable)**, and **don't allow duplicate keys**.

Example:

```
student = {
    "name": "Saumya Singh",
    "age": 25,
    "city": "Sultanpur"
}
```

Here:

- `"name"`, `"age"`, `"city"` → **keys**
- `"Saumya Singh"`, `21`, `"Delhi"` → **values**

Accessing Values

You can access a value using its key:

```
print(student["name"])     # Saumya Singh
print(student["city"])     # Delhi
```

Adding or Updating Values

You can add new key-value pairs or modify existing ones:

```
student["college"] = "ABC University"   # add new key-value
```

```
student["age"] = 22                    # update existing value
print(student)
```

Removing Items

```
student.pop("city")      # removes key 'city'
print(student)
```

Dictionary Methods

Method	Description	Example
`.keys()`	Returns all keys	`student.keys()`
`.values()`	Returns all values	`student.values()`
`.items()`	Returns all key-value pairs as tuples	`student.items()`
`.get(key)`	Returns value of a key safely	`student.get("name")`
`.update(new_dict)`	Updates dictionary with another	`student.update({"city": "Lucknow"})`

Nested Dictionary

You can store another dictionary inside a dictionary.

Example:

```
profile = {
    "username": "saumya1singh",
    "details": {
        "followers": 1200,
        "verified": True
    }
}
```

```
print(profile["details"]["followers"])   # 1200
```

Practice Question 1

Create a dictionary named `marks` to store marks of 3 subjects.
Add the subjects one by one and print the final dictionary.

Example:

```
Input:
Maths → 90
Science → 85
English → 88

Output:
{'Maths': 90, 'Science': 85, 'English': 88}
```

2️⃣ Sets in Python

Definition

A **set** is a collection of **unordered and unique items**. Sets automatically remove duplicate elements and are written using **curly braces `{ }`**.

Example:

```
languages = {"Python", "Java", "C++", "Python"}
print(languages)
# Output: {'C++', 'Java', 'Python'}
```

Creating a Set

```
empty_set = set()      # Empty set
nums = {1, 2, 3, 4}    # Non-empty set
```

Set Properties

- Unordered → no fixed index positions
- Unique → no duplicates
- Mutable → elements can be added or removed
- Cannot contain mutable elements like lists or dictionaries

Adding and Removing Elements

```
nums = {1, 2, 3}
nums.add(4)
nums.remove(2)
print(nums)   # {1, 3, 4}
```

Other Useful Methods

Method	Description	Example
`.add(el)`	Adds an element	`nums.add(5)`
`.remove(el)`	Removes element	`nums.remove(1)`
`.clear()`	Empties the set	`nums.clear()`
`.pop()`	Removes a random element	`nums.pop()`
`.union(set2)`	Combines both sets	`{1,2}.union({2,3}) → {1,2,3}`
`.intersection(set2)`	Common elements of both sets	`{1,2,3}.intersection({2,3,4}) → {2,3}`

Practice Question 2

You are given a list of programming languages:
`["Python", "Java", "C++", "Python", "Java", "C"]`
Convert it into a set and print how many **unique languages** Divya knows.

Expected Output:

```
{'Python', 'Java', 'C++', 'C'}
Divya knows 4 unique languages.
```

3️⃣ Difference Between Dictionary and Set

Feature	Dictionary	Set
Structure	Stores data as key–value pairs	Stores unique values only
Syntax	`{ "key": value }`	`{value1, value2, ...}`
Mutable	Yes	Yes
Duplicates	Keys are unique	All elements unique
Indexing	Not supported	Not supported

✅ Summary

- **Dictionary:** Stores key–value pairs, mutable, unique keys.
- **Set:** Stores unique, unordered elements.
- Both are very useful for **data handling and searching operations**.

Mini Assignment

1. Create a dictionary storing meanings of 3 English words.
2. Create a set of numbers and show union and intersection with another set.

3. Try to add both integer `9` and float `9.0` to a set and observe what happens.
   (Hint: You can convert one into a string to make both unique.)